

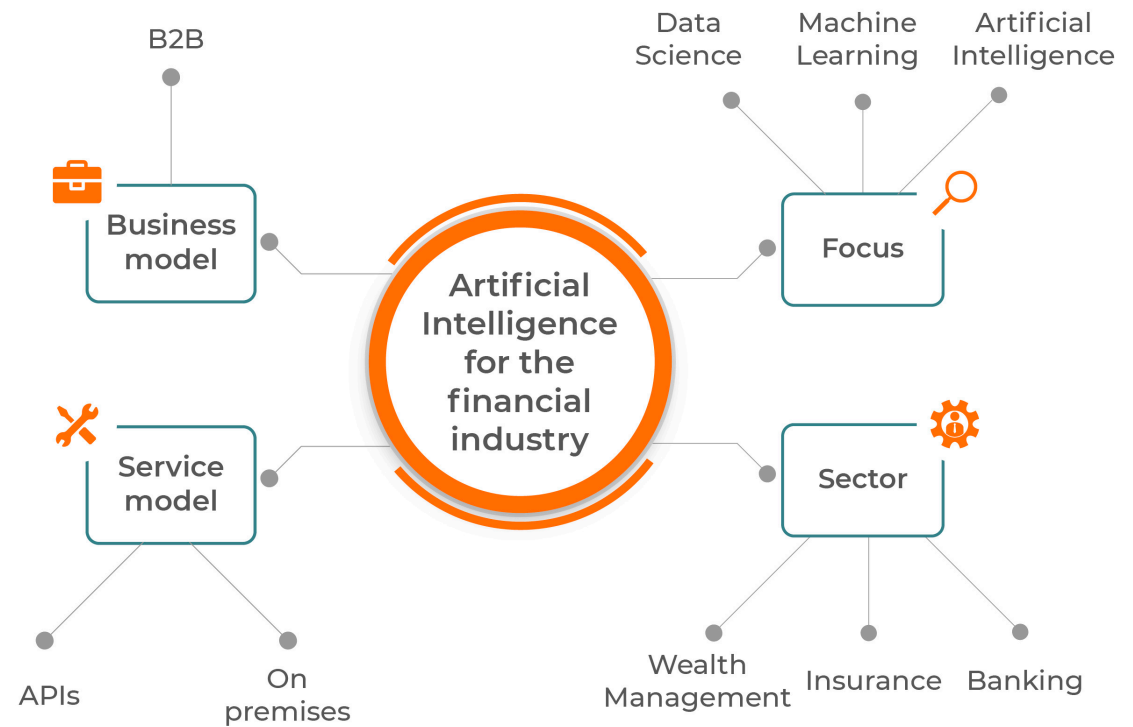
Module #1

Data-driven customer profiling for financial institutions/Fintechs

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Why here?
Because of
Wealthype-ai SpA



Overall goals

- You will savor the practice of Machine Learning in the Fintech sector with a series of business cases
- We will cover various aspects of the business (e.g., customer intelligence, digital marketing, investing, risk analysis) and a significant selection Data Science techniques (obviously not all)
- I would like to stimulate your ability to solve real problems with Data Science - having modeling vision – not just writing a piece of code, or using that new fancy model...





Organizational aspects



GROUP WORK & PRESENTATIONS

Group formation & regular assignments

- **Groups** of 4-5 students
- Each group is expected to work on all **4 business cases** - there are penalties for not working on cases (see course guidelines)
- Output required: **presentation** (pdf) **plus code** (e.g., Python code you should be able to show) + **optional Extended Abstract** (max 3 pages)
- This material must be **submitted at least 3 business days before the discussion** (e.g., for a discussion on Friday, it must be submitted by 8 pm on the previous Tuesday)

Presentation & Scoring

- Each business case will be **rated on a scale of 0-20**
- On the day of the discussion, **I will select 5 groups to present their work** (these can be the top 5, or groups whose work is representative)
- **The presentation will be considered for evaluation** - at least **2 members** per group must be present, otherwise the evaluation will be negative
 - **Final grade = average of 4 assessment**
 - If 3 business cases are submitted → final grade = average – 2 points
 - If 2 business cases are submitted → final grade = average – 4 points
 - If 1 business cases is submitted → End-of-the-Course Exam
- Each successful work/presentation earns points for ALL group members
- **Mentorship sessions** will be scheduled before poster sessions – they are videocall with me, if you have questions, you need some coaching, brainstorming, etc → **just email me here: raffaele.zenti@wealthype.it**

Hands on lessons

A number of different business cases =
problems to solve with Data Science

- I present the case from a **business point of view**
- I present an associated **dataset**
- We will do some **brainstorming**: I want to see you come up with ideas on how to solve the problem
- **I show you some code** that solves the problem: My code is never “definitive”, it is just a starting point, and it requires work on your part - but you will find a lot of hints
- I leave you to **develop your own ideas and code** (the aim of the course), **playing at home, in groups**
- I repeat - there are **ancillary hours («office hours»)** of my mentoring to support you in your work: just drop me an **email** here:
raffaele.zenti@wealthype.it



Coding

- In my lectures I will use Python, sometimes just example/code snippets
- You can use **the programming language of your choice**, Python, Matlab, or R
- **Use markdowns, comments, formatting and make the code as clear as possible** (better if you use notebooks like Colab, Jupyter, etc)

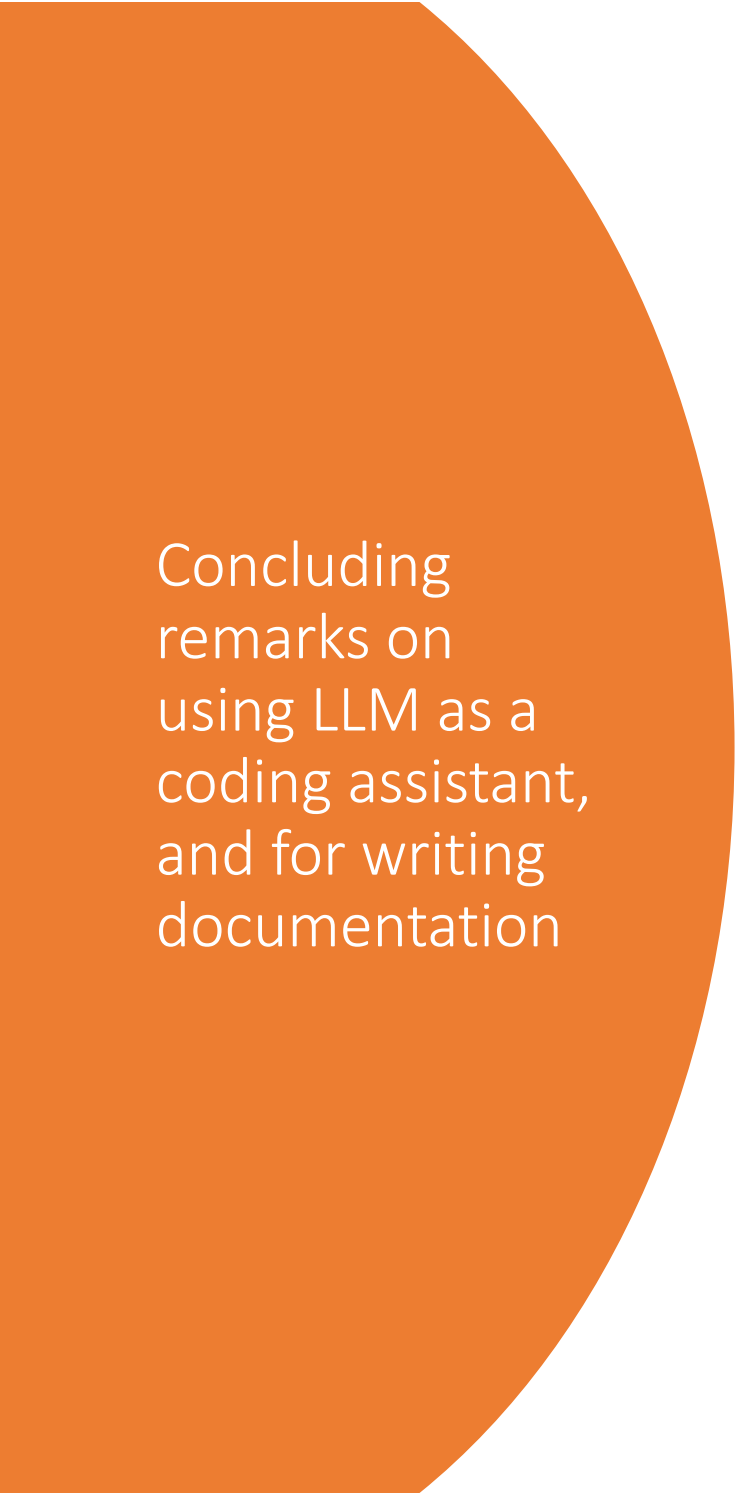
LEVERAGING LLMs FOR CODING SUPPORT – My 2 Cents

- **Encouraged use:** LLMs can significantly enhance your coding efficiency and problem-solving capabilities
«The most popular programming language in 2026: English»

- **Develop Cointelligence:** Build a collaborative relationship with AI where
 - YOU maintain strategic direction and control
 - YOU define the approach and goals clearly—the strategy must come from you, not the AI
 - YOU govern the development process and retain oversight of all generated solutions
 - YOUR critical judgment is key
 - The LLM provides suggestions, implementations (like having a smart assistant), and alternatives, becoming a great partner

- **Human-in-the-Loop Value:**

"Use AI as a powerful tool, not as a replacement for your own understanding."

A large orange shape on the left side of the slide, consisting of a vertical rectangle on the left and a quarter-circle on the right.

Concluding
remarks on
using LLM as a
coding assistant,
and for writing
documentation

You'll be asked to explain what you did, why, what the advantages and disadvantages were, and what critical issues emerged.

If you delegate everything to AI, you're screwed. In life, I mean.

Furthermore, I'm not necessarily going to call on the best groups to present here in front of everyone; I might even call on less than convincing work. Which I might suspect is entirely the product of AI.

Homework: a note



It is designed to familiarize you with real problems, to let you try different techniques



It is not a competition to see who has the best model accuracy, or MSE, or whatever...



You don't have to get better results than I have shown in class, you have to work properly



It's just some experimenting with the data, your brain, your team, your LLM assistant

CREATING AN EFFECTIVE PRESENTATION: KEY PRINCIPLES



Structure matters: Organize content logically

Clear sections with consistent layout



Visual hierarchy: Direct attention through font sizing and positioning

Headings
Body text



Show, don't tell: Aim for 40% text, 60% visuals

Replace text with diagrams, charts, and images



Simplify & Synthesize: Be ruthlessly concise

Focus only on key messages and findings, time is short
10÷15 slides



Presentation skills: Be prepared to present your work in front of the class

Practice

PROJECT WORK EVALUATION CRITERIA

Deliverables

- Concise **presentation** (10÷15 slides)
- Well-organized **code** (Python, R, Matlab, or mixed)
- OPTIONAL: Extended Abstract (max 3 pages)

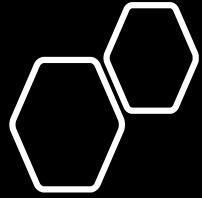
Evaluation Framework

- **Modeling vision & Problem-solving (★★)**
 - Ability to "connect the dots" in practical scenarios
 - Focus on solving the actual business problem
 - Pragmatic approach over unnecessary complexity
 - Clear, solid logic
- **Code quality (★★)**
 - Clean, very readable (generous use of markdowns), and well-structured code
 - Intelligent use of functions and libraries
 - Effective integration if using multiple languages
- **Presentation skills (★★)**
 - Clear, concise communication of methods and results
 - Focus on core findings without excess detail
 - Logical flow and professional delivery
 - Balance of text and visuals
- **Originality & exploration (★★)**
 - Differentiation from provided examples - Novel approaches (from class "Hints" or even better your own ideas)
 - Exploration/contrasting alternative approaches
 - Occam's Razor: valuing simplicity when equally effective



Business case #1:

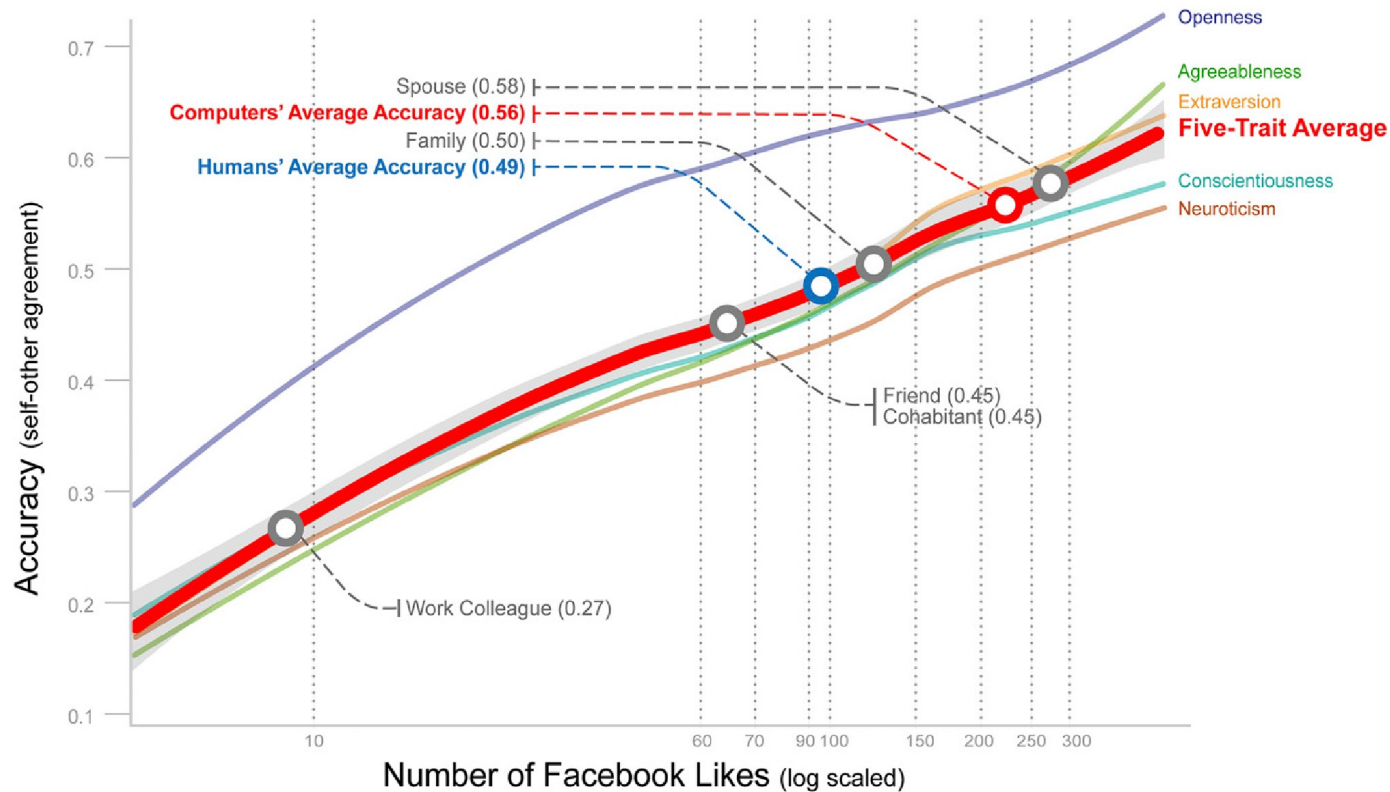
Segmenting Clients



AI/Machine Learning aimed for....?

- Understand or predict:
 1. Markets
 2. People (Customers) →
this is the case

Motivation: AI/ML and people



A Machine Learning algorithm needs 70, 150 and 300 Facebook Likes, respectively, to outperform an average friend, family member, and spouse in assessing your personality*

*Five-Factor Model model

Source: Youyou, W., Kosinski, M., & Stillwell, D. (2015). Computer-based personality judgments are more accurate than those made by humans. PNAS Proceedings of the National Academy of Sciences of the United States of America, 112(4), 1036-1040

Why segment? To efficiently customize any service





Client segmentation: the old school in finance

- Money, e.g.
 - AuM
 - insurance premiums
 - number of trades p.a.
 - ...
- Age
- A combination of 1. and 2.

*(Yeah. Boring. There is room for some improvement
- unless you think that money and/or age fully
define a human being)*

Personas: general idea

- Premise: we are all different, but with common traits
- Personas (or “Marketing Personas”, or Financial Personas, in Financial Services) = **human prototypes with similar traits**
- Depending on the perspective, human prototypes change – here we take the financial perspective
- Key to summarize information ↔ **dimensionality reduction**
- That is, instead of reasoning on, say, 300k customers, you can reason on 6 TYPES of customers – which is easier

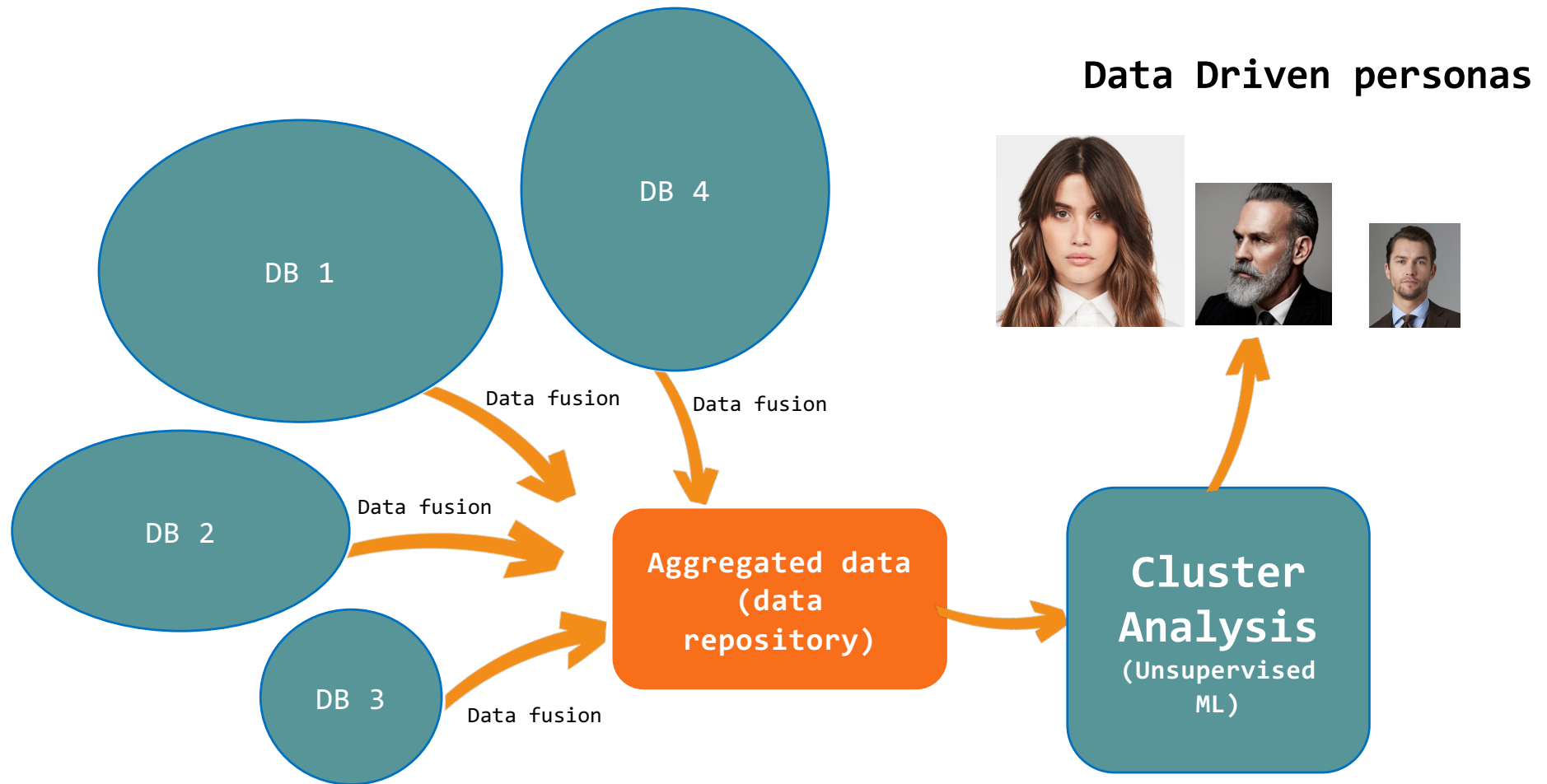


Personas according to traditional marketing

- Put some experts in a room
- Super-brainstorming
- Based on their experience, they will identify «customer prototypes», i.e., personas – they will likely include their biased personal ideas

(Better. But there is still room for some improvement)

Data-driven personas: (simplified) overview



The specific problem

- **Goal: segment bank clients based on the information contained in their corresponding vectors of features**
- Spoiler: in real world client data contain heterogeneous data (categorical nominal/ordinal, numerical ordinal/continuous)
- Introductory courses to unsupervised learning quite often discuss ideal use cases, such as tutorials using k-means, which works great, but only apply to numerical features – so we are out of the comfort zone...
- Now: let's inspect data and discuss!

A close-up, slightly angled view of a green calendar page. The page is divided into a grid of squares, each containing a number. The numbers are in a bold, sans-serif font. The numbers 6, 7, 11, 12, 13, 14, 15, 17, 18, 19, 20, 21, 22, 25, 26, 27, and 28 are in black. The numbers 16 and 23 are in a teal color. The text "Coding session starts" is overlaid in the center of the page in a white, sans-serif font. The text is slightly shadowed, giving it a 3D appearance as if it's floating above the calendar. The lighting is soft, and the overall tone is a muted green.

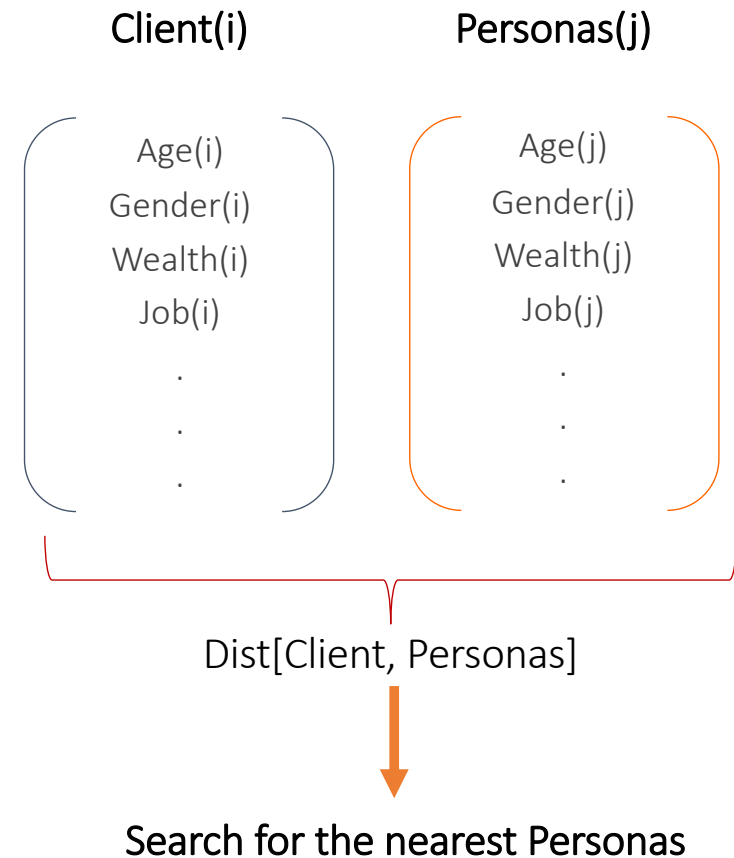
Coding session
starts

At the end of the process...

- Qualitative overlay might change the number of clusters, or might change centroids/medoids
- Each client is a vector:
 $z(i), i = 1, 2, 3, \dots n_{\text{Clients}}$
- Each Persona (centroid/medoid) is a vector:
 $C(j), j = 1, 2, 3, \dots n_{\text{Personas}}$
- Client(i) belongs to the closest Personas(j), i.e. the rule is:

$\text{ArgMin}\{ \text{dist}[z(i), C(j)] \}$

j



Real example – inside clusters (it's not your dataset)

Features	Cluster (2) = 21% “Wealthy widow”
Age	55-70
Gender	F
Job	Housewife, retired
Marital status	Widow, separated, divorced
Family	1
Financial education	Below average
Geographical area	Italy
Size of the municipality	20k÷50k
Income	Above average
Mortgage	N
Short term loans	N
Real estate wealth	Huge
Financial assets	Above average
Socio-demographic risk	Average
Geo-seismic risk	Above average
Digital propensity	Low



Service model:

- Physical
- Physical+Call center

Main needs:

- Long Term Care
- Inheritance
- Investments (low risk, capital protection)
- Premium credit cards

Real example – inside clusters

Features	Cluster (3) = 2% “Top jobs”
Age	50-65
Gender	M
Job	Manager, professional, business owner
Marital status	Married
Family	2-3
Financial education	High
Geographical area	North Italy, large cities
Size of the municipality	Above 200k
Income	High
Mortgage	Y
Short term loans	N
Real estate wealth	Above average
Financial assets	High
Socio-demographic risk	Above average
Geo-seismic risk	Low
Digital propensity	High



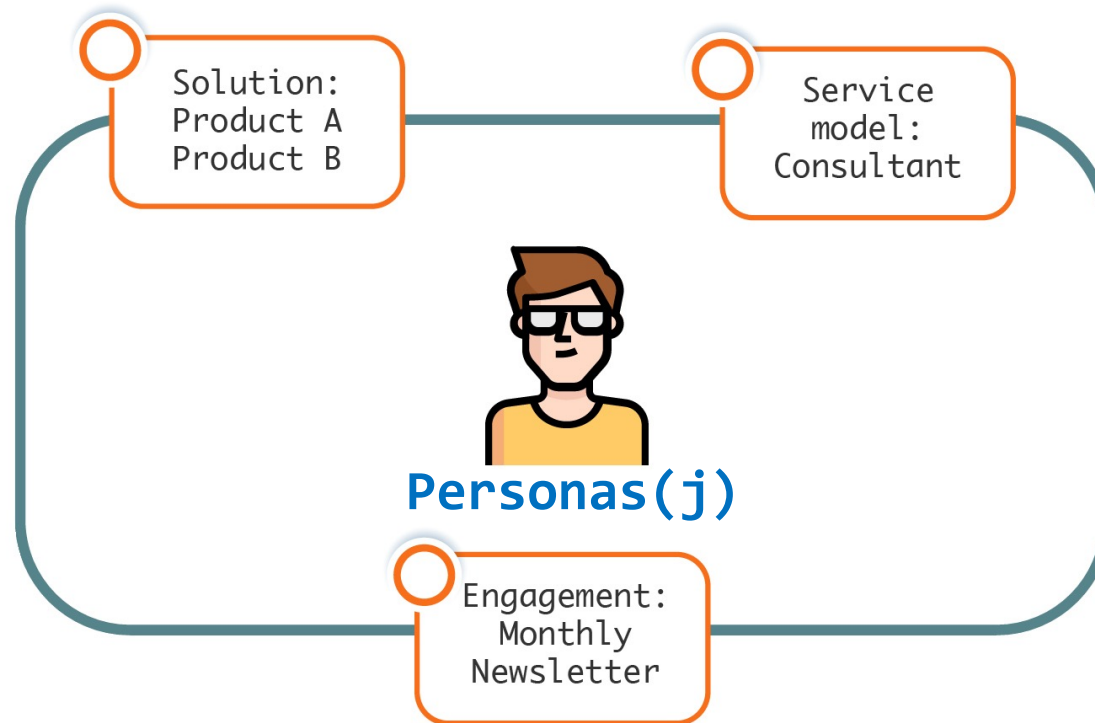
Service model:

- Digital
- Physical+call center

Main needs:

- Long Term Care
- Family/Home protection
- Death insurance
- Investments
- Premium credit cards

Using Personas to customize financial services in an industrial way



- Each Persona has her service/communication model, products, etc
- But Personas are typically 5÷15 → you reduce business complexity

Take home on data- driven personas

- Application of Unsupervised ML + qualitative overlay (strongly suggested)
- Much better segmentation, based on empirical evidence
- Several business applications:
 - customized products, services, channels
 - product governance and compliance (Retail Investment Strategy)
 - dedicated communication tools for each personas
 - precision marketing online/offline
 - data enrichment if we have a limited amount of information (a handful of features) → then you can start engaging customers
 - generating synthetic data, for ML training and simulations
 - ...

Now **YOU**

- **It's your turn: use your favorite techniques to segment customers, write the code and get your hands on that data...**

Next steps

- **Office Hours = mentorship**

Book if you want to talk to me; **write to me at this email address:**

raffaele.zenti@wealthype.it

(NOT the Politecnico email, pls)